

A Convention-Explicit Hopf Atlas for the Brusselator

Route-A source-local certificate HCS-C311

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Abstract

For every $A > 0$ and $B \geq 0$, we prove positive-quadrant global existence for the Brusselator and give its complete node, focus, defective, and Hopf equilibrium atlas.

1 Global existence and complete linear atlas

Consider

$$\dot{x} = A - (B + 1)x + x^2y, \quad \dot{y} = Bx - x^2y, \quad A > 0, \quad B \geq 0. \quad (1)$$

At $x = 0$ the first component equals $A > 0$, and at $y = 0$ the second is $Bx \geq 0$; hence the closed nonnegative quadrant is invariant. There $(x + y)' = A - x \leq A$, so neither coordinate can blow up in finite time. Every nonnegative solution is therefore global.

Theorem 1 (All-parameter equilibrium chambers). *Equation (1) has the unique nonnegative equilibrium $E = (A, B/A)$. Put $\tau = B - 1 - A^2$. Its Jacobian is*

$$J = \begin{pmatrix} B - 1 & A^2 \\ -B & -A^2 \end{pmatrix}, \quad \text{tr } J = \tau, \quad \det J = A^2. \quad (2)$$

Within $B \geq 0$, the increasing- B atlas is

range or boundary	type
$B < (A - 1)^2$	stable node
$B = (A - 1)^2$	defective stable node
$(A - 1)^2 < B < 1 + A^2$	stable focus
$B = 1 + A^2$	Hopf pair $\pm iA$
$1 + A^2 < B < (A + 1)^2$	unstable focus
$B = (A + 1)^2$	defective unstable node
$B > (A + 1)^2$	unstable node

Empty ranges are omitted; in particular $A = 1, B = 0$ is the stable defective boundary.

Proof. At equilibrium, adding the equations gives $x = A$, then the second gives $y = B/A$. Formula (2) follows by differentiation. Its eigenvalues are $\tau/2 \pm \sqrt{\tau^2 - 4A^2}/2$. The signs of τ and of this discriminant give every row. At equality the positive off-diagonal entry A^2 prevents J from being scalar, hence the repeated eigenvalue is defective. $\square$

Source lineage

References

- [1] I. Prigogine and R. Lefever, “Symmetry breaking instabilities in dissipative systems. II,” *J. Chem. Phys.* 48 (1968), 1695–1700. DOI: [10.1063/1.1668896](https://doi.org/10.1063/1.1668896).
- [2] Y. A. Kuznetsov, *Elements of Applied Bifurcation Theory*, 3rd ed., Springer, 2004. DOI: [10.1007/978-1-4757-3978-7](https://doi.org/10.1007/978-1-4757-3978-7).